10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 12.3 — Probability (Multiplication Law)

Unit 12 | Exercise 12.3 | New Book (PECTAA)

Exercise 12.3 (Solved)

This exercise is the multiplication law. With replacement the two events are independent and we simply multiply. Without replacement the second probability is taken from the smaller total.

1. Two numbers are randomly chosen from 1 to 10 with replacement. Find the probability that: (i) both are prime (ii) their product is even.

Total outcomes n(S) = 10 × 10 = 100

Primes between 1 and 10: {2, 3, 5, 7}, so 4 primes.

(i) P(first prime) = 4 / 10 and P(second prime) = 4 / 10

P(both prime) = 4 / 10 × 4 / 10 = 16 / 100 = 4 / 25

(ii) A product is even when at least one number is even.

P(odd) = 5 / 10 = 1 / 2 for each draw

P(both odd) = 1 / 2 × 1 / 2 = 1 / 4

P(product even) = 1 − 1 / 4 = 3 / 4

Answer: (i) 4/25 (ii) 3/4

2. One letter is chosen from the word PUNJAB and another from LAHORE. What is the probability that: (i) both are vowels (ii) one is consonant and other is vowel.

PUNJAB has 6 letters with vowels U, A, so 2 vowels and 4 consonants.

LAHORE has 6 letters with vowels A, O, E, so 3 vowels and 3 consonants.

n(S) = 6 × 6 = 36

(i) P(both vowels) = 2 / 6 × 3 / 6 = 6 / 36 = 1 / 6

(ii) One consonant and one vowel can happen in two ways:

vowel from PUNJAB, consonant from LAHORE = 2/6 × 3/6 = 6/36

consonant from PUNJAB, vowel from LAHORE = 4/6 × 3/6 = 12/36

P(one consonant and one vowel) = 6/36 + 12/36 = 18/36 = 1/2

Answer: (i) 1/6 (ii) 1/2

3. A single dice is rolled twice. Find the probability that one roll is a multiple of 3 and the other is a 5.

n(S) = 6 × 6 = 36

Multiples of 3 on one roll: 3, 6, so 2 outcomes.

The multiple of 3 may come on the first roll or on the second roll.

P = (2/6 × 1/6) + (1/6 × 2/6) = 2/36 + 2/36 = 4/36 = 1/9

Answer: P(one roll multiple of 3 and other 5) = 1/9

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4. Two dice are rolled. Find the probability of getting an odd number on one and a multiple of 2 on other.

n(S) = 36

Odd numbers are 1, 3, 5 (3 outcomes) and multiples of 2 are 2, 4, 6 (3 outcomes).

P = (3/6 × 3/6) + (3/6 × 3/6) = 9/36 + 9/36 = 18/36 = 1/2

Answer: P(odd on one and even on other) = 1/2

5. From a pack of well shuffled cards, two cards are drawn at random one by one with replacement. Find the probability that the first is heart and second is king.

There are 13 hearts and 4 kings in a pack of 52 cards.

With replacement the two draws are independent.

P(first heart) = 13 / 52 = 1 / 4

P(second king) = 4 / 52 = 1 / 13

P = 1 / 4 × 1 / 13 = 1 / 52

Answer: P(first heart and second king) = 1/52

6. If two cards are selected from a standard deck of 52 cards without replacement, find the probability that (i) Both are black (ii) Both are queens (iii) Both are spades (iv) Both are diamonds.

Without replacement, after the first card 51 cards are left.

(i) P(both black) = 26 / 52 × 25 / 51 = 650 / 2652 = 25 / 102

(ii) P(both queens) = 4 / 52 × 3 / 51 = 12 / 2652 = 1 / 221

(iii) P(both spades) = 13 / 52 × 12 / 51 = 156 / 2652 = 1 / 17

(iv) P(both diamonds) = 13 / 52 × 12 / 51 = 1 / 17

Answer: (i) 25/102 (ii) 1/221 (iii) 1/17 (iv) 1/17

7. Saleem draws two cards one by one without replacement from a well shuffled pack of 52 playing cards. What is the probability that first card is jack and the second card is queen.

P(first card is a jack) = 4 / 52 = 1 / 13

After taking one jack, 51 cards are left and all 4 queens are still there.

P(second card is a queen) = 4 / 51

P(jack then queen) = 1 / 13 × 4 / 51 = 4 / 663

Answer: P(first jack, second queen) = 4/663

Remember

With replacement: the total stays the same, so the events are independent.

Without replacement: subtract one card or ball before the second draw.

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